LESSON 2.4

Represent and Describe Data in a Box Plot

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can make and read a box plot to summarize a data set.

WHY THIS MATTERS The basketball league wants to compare scoring across teams. Here are the points scored by each player on the Wildcats in their last game: 4, 6, 8, 10, 12, 14, 15, 18, 22, 24, 28. The league needs a quick visual that shows the spread, the middle 50%, and any high or low scorers. Time to build a box plot!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Box Plot *Spanish: Diagrama de caja*	A graph that shows how data is spread using five key numbers.	Box with two whiskers
Median *Spanish: Mediana*	The middle number when you put them in order.	Data: 10, 15, 20, 25, 30 → median is 20 (the 3rd value)
Quartile *Spanish: Cuartil*	Numbers that split the data into four equal parts.	Q1, median, Q3
Interquartile Range *Spanish: Rango intercuartílico*	The spread of the middle half of the data (Q3 − Q1).	If Q1 = 12 and Q3 = 20, then IQR = 20 − 12 = 8 — the middle 50% spans 8 units
Data distribution *Spanish: Distribución de datos*	How the data looks: where it sits and how spread out it is.	A box plot where the median is centered in the box shows symmetric distribution
Variability *Spanish: Variabilidad*	How spread out the numbers are.	Small IQR = low variability in the middle 50%. Large IQR = high variability

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Times (min): 14, 5, 20, 9, 23, 11, 16.

1. Order: 5, 9, 11, 14, 16, 20, 23. Median = 14.
2. Lower half 5, 9, 11: Q1 = 9. Upper half 16, 20, 23: Q3 = 20.
3. Box from 9 to 20, line at 14, whiskers to 5 and 23.

Answer: 5, 9, 14, 20, 23

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New, smaller data set: 2, 5, 5, 8, 11. First, what is the median (the middle value)?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In order the values are _____ .
- The median is _____ , so the lower half is _____ .
- Quartile 1 is _____ and quartile 3 is _____ .
- The box stretches from _____ to _____ , which holds _____ of the data.

Word bank box plot | quartile | median | minimum | maximum | whisker | box | order | lower half | upper half | middle | spread

WATCH OUT!

A common mistake

A common mistake in Display Data: Box Plots is treating a quartile as just one data point instead of the median of a half

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

On a box plot, what does the line inside the box represent?

(A) The median
(B) The mean
(C) The mode
(D) The range

STEPS TO TRY

1. Picture a box plot: a box with whiskers, and a line cutting through the box. That line marks the CENTER of the data.
2. The box runs from Q1 to Q3, so the line inside splits the middle 50% in half. Think about which measure splits ordered data into halves.
3. First step: recall which measure means 'the middle value of the ordered data.' The line inside the box always marks that exact measure.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A box plot shows: Min = 10, Q1 = 15, Median = 20, Q3 = 28, Max = 35. What is the interquartile range (IQR)?

(A) 10
(B) 13
(C) 20
(D) 25

STEPS TO TRY

1. You're given five numbers, but IQR only uses two of them: Q1 and Q3. Ignore the min, median, and max.
2. IQR = Q3 − Q1. Find Q3 (28) and Q1 (15) in the summary — the IQR is their difference, the width of the box.
3. Set up the subtraction Q3 − Q1: that's 28 − 15. The difference you get is the IQR.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

3 **Circle the letter of the best answer. Show how you know.**

What percentage of data falls between Q1 and Q3 on a box plot?

(A) 25%
(B) 50%
(C) 75%
(D) 100%

STEPS TO TRY

1. Quartiles cut the data into 4 equal parts — 'quart' like 'quarter.' Each part holds the same share of the data.
2. Each quarter of the data is 25%. Count how many quarters sit between Q1 and Q3: Q1 to the median, then the median to Q3.
3. First step: Q1 to the median is one quarter — 25%. Add the quarter from the median up to Q3 to get the box's total share.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

4 **Circle the letter of the best answer. Show how you know.**

Find the range of: 14, 22, 8, 31, 17

(A) 8
(B) 18
(C) 23
(D) 31

STEPS TO TRY

1. Range = how far the data stretches: the gap between the biggest and smallest values only. The middle values don't matter.
2. Scan the list 14, 22, 8, 31, 17: find the maximum and the minimum, then subtract the min from the max.
3. First step: the max is 31 and the min is 8. Set up 31 − 8 — that difference is the range.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

EXPLAIN YOUR THINKING

The IQR is Q3 − Q1 = 26 − 10 = 16. What does this tell you about the middle 50% of scorers?

SENTENCE STARTER The middle 50% of players scored between ___ and ___ points. The IQR of ___ means the middle half of the data is spread across ___ points. This tells me ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help